

Reciprocity and Matrix Duality for Preorder Polytopes

Proofs of three conjectures

Interior translation, colored arrays, and generalized permutohedra

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Abstract

We investigate conjectures on the polytopes attached to finite preorders by Athanasiadis and Chapoton. The central observation is that subtracting the all-ones vector identifies the interior lattice points of their two-parameter polytope with the lattice points obtained by decreasing both parameters by one. Ordinary Ehrhart–Macdonald reciprocity along integer rays then proves their Conjecture 7.2. We establish an $(n + 1)$ -parameter strengthening and a second, falling-binomial formula for the lattice enumerator. This proves their Conjecture 4.4 on the h^* -polynomial and the ordinary, $q = 1$ specialization of Conjecture 4.9.

A separate construction replaces the two color classes of the preorder’s bipartite graph by independently prescribed numbers of copies. Postnikov’s lattice-point formula yields a heterogeneous array duality, which specializes to the matrix Ehrhart–zeta duality of Conjecture 4.2. Detailed proofs, examples, sign and boundary checks, and an exact-arithmetic verification program are provided. The program checks every labeled preorder on one through four elements. No conjectured shellability or real-rootedness is used.

Status. The results are proved in this manuscript relative to the two explicitly stated classical theorems. The work has not been externally reviewed or formally verified. The selected statements are conjectures in the public version of arXiv:2605.26916 examined on September 19, 2026; the literature search does not establish priority against unpublished work.

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1 The selected problem and the outcome

A finite preorder is a reflexive, transitive relation. Antisymmetry is not required, so equivalence classes may contain several elements. The operation of reversing this relation is a natural order-theoretic duality. The question studied here is how that duality interacts with integer vectors constrained by order ideals.

The starting point is Conjecture 7.2 of Athanasiadis and Chapoton [1], in the first public version dated May 26, 2026. For a preorder τ on an n -element set E , write

$$Q_\tau(r, s) = \left\{ \mathbf{x} \in \mathbb{R}_{\geq 0}^E : x(I) \leq r|I| + s \text{ for every nonempty order ideal } I \right\}, \quad x(I) := \sum_{e \in I} x_e. \quad (1)$$

For $r, s \in \mathbb{N}$, its lattice-point count is a polynomial $\mathbb{E}_\tau(r, s)$ of total degree n . Their conjecture asks whether

$$\boxed{\mathbb{E}_\tau(-u, -v) = (-1)^n \mathbb{E}_\tau(u-1, v-1).} \quad (2)$$

Here and throughout, $\mathbb{N} = \{0, 1, 2, \dots\}$. Negative arguments refer to *polynomial evaluation*, not to negative dilations interpreted as feasible sets.

1.1 What is proved

The main results have two independent mechanisms. Interior translation proves (2) and its fully weighted extension. A bipartite graph with independently replicated vertices proves the matrix duality. A binomial specialization of the first result yields the word formula.

Statement in [1]	Result here	Location
Conjecture 7.2	Double Ehrhart reciprocity, with a stronger multi-variate identity	Section 4
Conjecture 4.2	Matrix Ehrhart–zeta duality, with heterogeneous copy counts	Section 6
Conjecture 4.4	Descent interpretation of the h^* -polynomial	Section 5
Equation (22) in Conjecture 4.9	Ordinary zeta evaluation at -1	Corollary 5.4

The full q -identity in Conjecture 4.9 is *not* proved here. Nor do we claim to settle the paper’s questions about shellability, real-rootedness, γ -positivity, or support enumerators. Three conjectures are proved, but one of the three is a consequence of another; these are not three unrelated breakthroughs.

1.2 The shortest proof of the original target

For positive integers r, s , the map

$$\mathbf{x} \mapsto \mathbf{x} - \mathbf{1}$$

is a bijection

$$\text{int } Q_\tau(r, s) \cap \mathbb{Z}^E \longleftrightarrow Q_\tau(r-1, s-1) \cap \mathbb{Z}^E. \quad (3)$$

Indeed, integrality changes each strict upper inequality into $x(I) \leq r|I| + s - 1$, and subtracting one in each coordinate removes $|I|$ from its left-hand side. Coordinate positivity becomes nonnegativity. Apply ordinary Ehrhart reciprocity to the *fixed* lattice polytope $Q_\tau(r, s)$, using $tQ_\tau(r, s) = Q_\tau(tr, ts)$. This proves (2) on all positive integer pairs, hence as a polynomial identity. The rest of the article verifies the hypotheses, extends the argument, and develops its consequences.

1.3 Provenance and the scope of the novelty claim

The version history and the statements of Conjectures 4.2, 4.4, and 7.2 in [1] were checked directly, including the rendered PDF pages. Focused public searches did not locate a resolution. This supports selecting them as research targets; it does not certify the absence of private or unindexed proofs. The underlying counting and reciprocity tools are classical. The contribution of this manuscript is the specialization, the explicit translation and cloning constructions, and the resulting proofs and extensions.

2 Notation and the two classical inputs

2.1 Preorders, ideals, and finite downsets

Fix a nonempty finite set E , with $n = |E|$, and a preorder $\leq_\tau$. Its dual τ^* is defined by $e \leq_{\tau^*} f$ if and only if $f \leq_\tau e$. An order ideal is a subset closed downwards. An order filter is closed upwards. Write $\mathcal{J}(\tau)$ for the family of all order ideals, including the empty ideal. In displayed polytope inequalities we use only the nonempty ideals.

For $S \subseteq E$, let

$$D_\tau(S) = \{e : e \leq_\tau f \text{ for some } f \in S\}, \quad U_\tau(S) = \{f : e \leq_\tau f \text{ for some } e \in S\}.$$

Abbreviate $U_\tau(\{e\})$ to $U_\tau(e)$. Reflexivity and transitivity ensure that these are respectively the ideal and filter generated by S . For any vector $\mathbf{b}$, let $b(S) = \sum_{e \in S} b_e$ and $|\mathbf{b}| = b(E)$.

Define

$$Q_\tau := Q_\tau(1, 0), \quad P_\tau := Q_\tau \cap \mathbb{Z}^E, \quad B_\tau := \{\mathbf{a} \in P_\tau : |\mathbf{a}| = n\}. \quad (4)$$

The order on P_τ is coordinatewise comparison. It is a finite downset in $\mathbb{N}^E$. In particular, if $\mathbf{a} \in P_\tau$, then the interval $[0, \mathbf{a}]$ is the product of chains $\prod_{e \in E} \{0, 1, \dots, a_e\}$.

Lemma 2.1. *The maximal elements of P_τ are precisely B_τ .*

Proof. An element of coordinate sum n cannot be increased because the ideal E provides the bound $x(E) \leq n$. Conversely, suppose that $\mathbf{a} \in P_\tau$ has $|\mathbf{a}| < n$ and cannot be increased in any coordinate. For each $e \in E$, failure of $\mathbf{a} + \mathbf{e}_e$ to be feasible gives an ideal I_e containing e with $a(I_e) = |I_e|$. Such an ideal is called tight.

The union of two tight ideals is tight. To see this, use additivity to write

$$a(I \cup J) + a(I \cap J) = a(I) + a(J) = |I| + |J| = |I \cup J| + |I \cap J|.$$

The two summands on the left are each at most the corresponding cardinalities on the right. Equality of the sums forces equality for both summands. This also covers the empty intersection. Repeating the union operation shows that $\bigcup_{e \in E} I_e = E$ is tight, contradicting $|\mathbf{a}| < n$. Thus some coordinate can be increased, as required. $\square$

2.2 Ehrhart–Macdonald reciprocity

If $K \subset \mathbb{R}^d$ is a full-dimensional lattice polytope, its Ehrhart polynomial $L_K(t)$ satisfies

$$L_K(t) = \#(tK \cap \mathbb{Z}^d) \quad (t \in \mathbb{N}), \quad L_K(-t) = (-1)^d \#(\text{int}(tK) \cap \mathbb{Z}^d) \quad (t \geq 1). \quad (5)$$

We use this classical theorem in the form developed in Stanley’s treatment of Ehrhart–Macdonald reciprocity [3, Section 6]. We do not need a separate multivariate reciprocity theorem: restricting a polynomial to each positive integer ray is sufficient.

For a full-dimensional n -dimensional lattice polytope K , define its h^* -polynomial by

$$\sum_{m \geq 0} L_K(m) t^m = \frac{h^*(K, t)}{(1-t)^{n+1}}. \quad (6)$$

The finite-difference relation implicit in this definition will be enough for the computations below.

2.3 Postnikov’s lattice-point formula

This is the other substantive imported theorem. Let R be a finite set of size $d+1$, and let $I_0, I_1, \dots, I_m$ be nonempty subsets of R , with $I_0 = R$. Associate to them a bipartite graph with one left vertex for each I_i , right vertex set R , and adjacency given by membership. A vector $\alpha = (\alpha_0, \dots, \alpha_m) \in \mathbb{N}^{m+1}$ is *draconian* when

$$\sum_{i=0}^m \alpha_i = d, \quad \sum_{i \in A} \alpha_i \leq \left| \bigcup_{i \in A} I_i \right| - 1 \quad \text{for every nonempty } A \subseteq \{0, \dots, m\}. \quad (7)$$

Here $\Delta_I = \text{conv}\{\mathbf{e}_j : j \in I\}$ is a simplex in $\mathbb{R}^R$, without an additional origin.

Theorem 2.2 (Postnikov). *For nonnegative integers $y_0, \dots, y_m$,*

$$\# \left(\left(\sum_{i=0}^m y_i \Delta_{I_i} \right) \cap \mathbb{Z}^R \right) = \sum_{\alpha \text{ draconian}} \binom{y_0 + \alpha_0}{\alpha_0} \prod_{i=1}^m \binom{y_i + \alpha_i - 1}{\alpha_i}. \quad (8)$$

This is the untrimmed form of [2, Theorem 11.3], obtained by increasing by one the coefficient of the full simplex. It is also the form used in [1, Theorem 2.1]. Repeated neighborhoods are permitted, as are zero coefficients. The offset in the $i=0$ factor is essential.

All binomial coefficients in polynomial identities mean

$$\binom{z}{k} = \frac{z(z-1) \cdots (z-k+1)}{k!} \quad (k \in \mathbb{N}), \quad \binom{z}{0} = 1. \quad (9)$$

In particular, setting a coefficient $y_i = 0$ annihilates its factors with $\alpha_i > 0$ when $i \neq 0$, but setting $y_0 = 0$ does not annihilate the α_0 factor.

3 Weighted preorder polytopes

3.1 A Minkowski-sum description with independent capacities

For $S \subseteq E$, write $\Delta_S = \text{conv}(\{\mathbf{0}\} \cup \{\mathbf{e}_e : e \in S\})$ in $\mathbb{R}^E$; thus $\Delta_\emptyset = \{\mathbf{0}\}$. The bold simplex includes the origin, unlike Δ_I in Theorem 2.2.

For $\mathbf{b} \in \mathbb{R}_{\geq 0}^E$ and $s \geq 0$, define

$$\mathcal{Q}_\tau(\mathbf{b}; s) = \{\mathbf{x} \geq \mathbf{0} : x(I) \leq b(I) + s \text{ for all } \emptyset \neq I \in \mathcal{J}(\tau)\}. \quad (10)$$

The uniform family is $\mathcal{Q}_\tau(r, s) = \mathcal{Q}_\tau(r\mathbf{1}; s)$.

Proposition 3.1. *For arbitrary nonnegative real parameters,*

$$\mathcal{Q}_\tau(\mathbf{b}; s) = \sum_{e \in E} b_e \Delta_{U_\tau(e)} + s \Delta_E. \quad (11)$$

Consequently, this is a lattice polytope when the parameters are integers.

Proof. Denote the right-hand side by M . If I is an ideal, the summand indexed by e can contribute to a coordinate in I only if $e \in I$. Its total contribution to $x(I)$ is at most b_e . The last summand contributes at most s . Therefore $M \subseteq \mathcal{Q}_\tau(\mathbf{b}; s)$.

For the reverse containment we compare linear functionals. Both sets are closed under decreasing nonnegative coordinates. Hence it suffices to consider a functional with nonnegative coefficients $\mathbf{c}$; for a general functional, replace its coefficients by their positive parts and note that the support function of M does not change.

For $t > 0$ let $S_t = \{f \in E : c_f \geq t\}$. For a feasible point, whenever $S_t \neq \emptyset$,

$$x(S_t) \leq x(D_\tau(S_t)) \leq b(D_\tau(S_t)) + s.$$

Integrate over $0 < t \leq \max_f c_f$. The elementary layer-cake identity gives

$$\begin{aligned} \mathbf{c} \cdot \mathbf{x} &= \int_0^{\max_f c_f} x(S_t) dt \\ &\leq \sum_{e \in E} b_e \max_{f \in U_\tau(e)} c_f + s \max_{f \in E} c_f. \end{aligned}$$

The last expression is exactly the maximum of $\mathbf{c} \cdot \mathbf{x}$ on M , since each simplex summand can choose a coordinate attaining its own maximum. Thus no linear functional separates a feasible point from M ; compact convex separation proves the desired containment. Boundedness follows from the ideal E .

For integer parameters the right-hand side is a Minkowski sum of lattice polytopes, hence is the convex hull of integer sums of their vertices. It is therefore a lattice polytope. $\square$

Remark 3.2. At $s = 0$ some inequalities for disconnected ideals are sums of inequalities for their components. This is not a license to omit those ideals when $s > 0$: summing component bounds would add the slack s several times. We keep every nonempty ideal throughout.

3.2 The multivariate lattice enumerator

Introduce the graph with left and right copies of $E \sqcup \{0\}$ and neighborhoods

$$I_0 = E \sqcup \{0\}, \quad I_e = U_\tau(e) \sqcup \{0\} \quad (e \in E).$$

The projection dropping coordinate 0 identifies the corresponding Minkowski sum, with weights $y_0 = s$ and $y_e = b_e$, with (11). For integer parameters, its inverse on the fixed-sum hyperplane is $x_0 = |\mathbf{b}| + s - |\mathbf{x}|$, so the projection preserves the affine integer lattices.

The following identification is [1, Lemma 4.1]; we recall its proof to make the dual orientation explicit.

Lemma 3.3. *A vector $(a_0, (a_e)_{e \in E})$ is draconian for this graph if and only if $\mathbf{a} \in P_{\tau^*}$ and $a_0 = n - |\mathbf{a}|$.*

Proof. The total-sum condition gives the asserted value of a_0 . A subset of left vertices containing 0 has all $n + 1$ right vertices as neighbors, so its inequality follows from nonnegativity and the total sum n . For a nonempty subset $A \subseteq E$, the neighborhood inequality is

$$a(A) \leq |U_{\tau}(A)|.$$

If A itself is a filter, this is $a(A) \leq |A|$. Conversely, the inequalities for all filters imply the inequality for an arbitrary A , since $a(A) \leq a(U_{\tau}(A)) \leq |U_{\tau}(A)|$. Filters of τ are ideals of τ^* . The condition for the filter E also guarantees $a_0 \geq 0$. $\square$

Proposition 3.4. *There is a polynomial $\mathbb{F}_{\tau}(\mathbf{b}, s)$ of total degree n such that*

$$\mathbb{F}_{\tau}(\mathbf{b}, s) = \#(Q_{\tau}(\mathbf{b}; s) \cap \mathbb{Z}^E) \quad (\mathbf{b} \in \mathbb{N}^E, s \in \mathbb{N}).$$

It is given by

$$\mathbb{F}_{\tau}(\mathbf{b}, s) = \sum_{\mathbf{a} \in P_{\tau^*}} \binom{s + n - |\mathbf{a}|}{n - |\mathbf{a}|} \prod_{e \in E} \binom{b_e + a_e - 1}{a_e}. \quad (12)$$

In particular, $\mathbb{E}_{\tau}(u, v) = \mathbb{F}_{\tau}(u\mathbf{1}, v)$.

Proof. Apply Theorem 2.2 and Lemma 3.3. Every summand has total degree at most n . The degree is exactly n , because setting $\mathbf{b} = \mathbf{0}$ leaves $\mathbb{F}_{\tau}(\mathbf{0}, s) = \binom{s+n}{n}$, the lattice count of $s\Delta_E$. Uniqueness follows because polynomials agreeing on $\mathbb{N}^{n+1}$ agree identically; this can be proved one variable at a time. $\square$

4 Reciprocity by interior translation

Lemma 4.1 (Interior translation). *Suppose $b_e \geq 1$ for every $e \in E$ and $s \geq 1$, all integers. Then $Q_{\tau}(\mathbf{b}; s)$ is full dimensional, and translation by $-\mathbf{1}$ gives a bijection*

$$\text{int } Q_{\tau}(\mathbf{b}; s) \cap \mathbb{Z}^E \longleftrightarrow Q_{\tau}(\mathbf{b} - \mathbf{1}; s - 1) \cap \mathbb{Z}^E. \quad (13)$$

Proof. The point $\mathbf{1}$ satisfies all coordinate inequalities strictly and all nonempty ideal inequalities strictly, since $|I| < b(I) + s$. There are finitely many inequalities, so a sufficiently small Euclidean ball about $\mathbf{1}$ lies in the polytope. This proves full dimension.

For a full-dimensional polytope described by finitely many inequalities with nonzero normals, a point is interior exactly when every defining inequality is strict. Redundant inequalities cause no problem: equality in any valid inequality with nonzero normal prevents a full ball around the point. Thus an integer point $\mathbf{x}$ is interior if and only if

$$x_e \geq 1 \quad (e \in E), \quad x(I) \leq b(I) + s - 1 \quad (\emptyset \neq I \in \mathcal{J}(\tau)).$$

Put $\mathbf{y} = \mathbf{x} - \mathbf{1}$. The coordinate conditions become $y_e \geq 0$, and

$$y(I) \leq b(I) - |I| + s - 1 = (\mathbf{b} - \mathbf{1})(I) + (s - 1).$$

These are exactly the inequalities on the right of (13). All transformations are reversible over $\mathbb{Z}^E$. $\square$

Theorem 4.2 (Multivariate reciprocity). *For every finite preorder on $n \geq 1$ elements,*

$$\boxed{\mathbb{F}_\tau(-\mathbf{b}, -s) = (-1)^n \mathbb{F}_\tau(\mathbf{b} - \mathbf{1}, s - 1)} \quad (14)$$

as an identity in $\mathbb{Q}[(b_e)_{e \in E}, s]$.

Proof. First fix positive integer parameters $\mathbf{b}, s$. The polytope $K = \mathcal{Q}_\tau(\mathbf{b}; s)$ is a full-dimensional lattice polytope by [Proposition 3.1](#) and [lemma 4.1](#). For every nonnegative integer t ,

$$tK = \mathcal{Q}_\tau(t\mathbf{b}; ts), \quad L_K(t) = \mathbb{F}_\tau(t\mathbf{b}, ts).$$

Both sides of the second equality are polynomials in t , so they agree also at $t = -1$. Ehrhart–Macdonald reciprocity therefore gives

$$\begin{aligned} \mathbb{F}_\tau(-\mathbf{b}, -s) &= L_K(-1) \\ &= (-1)^n \#(\text{int } K \cap \mathbb{Z}^E) \\ &= (-1)^n \mathbb{F}_\tau(\mathbf{b} - \mathbf{1}, s - 1), \end{aligned}$$

where the last equality is [Lemma 4.1](#).

The difference between the two sides of (14) is a polynomial vanishing on every tuple of positive integers. Fix all variables but one: the remaining univariate polynomial has infinitely many roots. Repeat this observation for the coefficient polynomials in the remaining variables. The difference is zero identically. $\square$

Corollary 4.3 (Conjecture 7.2 of [1]). *For all indeterminates u, v ,*

$$\mathbb{E}_\tau(-u, -v) = (-1)^n \mathbb{E}_\tau(u - 1, v - 1).$$

Proof. Set $b_e = u$ for every e in [Theorem 4.2](#). $\square$

4.1 Why the polynomial step matters

We intentionally proved the lattice bijection only when every parameter is positive. For example, $\mathcal{Q}_\tau(\mathbf{0}; 0)$ is a point, not a full-dimensional polytope. The identity at zero or negative parameters follows from the polynomial continuation, not from applying an interior formula to a lower-dimensional or nonexistent feasible set. Similarly, no assertion about the interior of a Minkowski sum being a sum of interiors is needed. The direct inequalities avoid that issue entirely.

Corollary 4.4 (Centered parity). *Define*

$$H_\tau(\mathbf{b}, s) = \mathbb{F}_\tau(\mathbf{b} - \tfrac{1}{2}\mathbf{1}, s - \tfrac{1}{2}).$$

Then $H_\tau(-\mathbf{b}, -s) = (-1)^n H_\tau(\mathbf{b}, s)$. Consequently, only monomials whose total degree has the same parity as n occur in H_τ . For the double polynomial, if $E_j(u, v)$ is its homogeneous part of degree j , then

$$E_{n-1}(u, v) = \tfrac{1}{2}(\partial_u + \partial_v)E_n(u, v). \quad (15)$$

Proof. Substitute $\mathbf{b} + \tfrac{1}{2}\mathbf{1}, s + \tfrac{1}{2}$ into (14). A monomial of degree j changes by $(-1)^j$ under simultaneous negation. For the last identity, write $\mathbb{E}_\tau(u, v) = H(u + \tfrac{1}{2}, v + \tfrac{1}{2})$, where H has no homogeneous part of degree $n - 1$. The degree- $n - 1$ terms come solely from translating its degree- n part, giving (15). $\square$

5 A second binomial formula and the word conjecture

Athanasiadis and Chapoton already explain in [1, Remark 7.3] that Conjecture 7.2 implies their word conjecture, and in [1, Remark 4.10(a)] that the word conjecture implies the ordinary zeta evaluation. We include these deductions in full, with an independent proof of the elementary word identity. The multivariate falling-binomial formula below strengthens the uniform specialization used there.

5.1 The falling-binomial form

Theorem 5.1. *For every finite preorder,*

$$\mathbb{F}_\tau(\mathbf{b}, s) = \sum_{\mathbf{a} \in P_{\tau^*}} \binom{s}{n - |\mathbf{a}|} \prod_{e \in E} \binom{b_e + 1}{a_e}. \quad (16)$$

In particular,

$$\mathbb{F}_\tau(\mathbf{b}, 0) = \sum_{\mathbf{a} \in B_{\tau^*}} \prod_{e \in E} \binom{b_e + 1}{a_e}, \quad L_{Q_\tau}(m) = \sum_{\mathbf{a} \in B_{\tau^*}} \prod_{e \in E} \binom{m + 1}{a_e}. \quad (17)$$

Proof. By Theorem 4.2, $\mathbb{F}_\tau(\mathbf{b}, s) = (-1)^n \mathbb{F}_\tau(-\mathbf{b} - \mathbf{1}, -s - 1)$. Apply (12) to the latter expression. Put $a_0 = n - |\mathbf{a}|$. The polynomial identities

$$\binom{-s - 1 + a_0}{a_0} = (-1)^{a_0} \binom{s}{a_0}, \quad \binom{-b_e + a_e - 2}{a_e} = (-1)^{a_e} \binom{b_e + 1}{a_e}$$

show that the sign of each summand is $(-1)^{a_0 + \sum_e a_e} = (-1)^n$. It cancels the external sign. This yields (16). At $s = 0$, the factor $\binom{0}{a_0}$ is nonzero exactly when $a_0 = 0$. The surviving vectors form B_{τ^*} by Lemma 2.1. $\square$

For nonnegative parameters, (16) has a simple counting interpretation. Take disjoint sets of sizes s and $b_e + 1$ for $e \in E$. Choose a total of n objects, with the vector of choices from the E -indexed sets lying in P_{τ^*} . Their number is $\mathbb{F}_\tau(\mathbf{b}, s)$. This interpretation counts sets without repetitions, whereas (12) uses selections with repetitions. The equality is an enumerative consequence of reciprocity; we do not assert a canonical bijection between these two models.

5.2 A self-contained multiset-word identity

Fix any total order on E , independently of the preorder. If $|\mathbf{a}| = n$, let $\mathfrak{S}_{\mathbf{a}}$ be the words of content $\mathbf{a}$. For $w = w_1 \cdots w_n$ define $\text{des}(w) = \#\{i \in \{1, \dots, n-1\} : w_i > w_{i+1}\}$.

Lemma 5.2. *For any $\mathbf{a} \in \mathbb{N}^E$ of coordinate sum $n \geq 1$,*

$$\sum_{m \geq 0} \left(\prod_{e \in E} \binom{m + 1}{a_e} \right) t^m = \frac{\sum_{w \in \mathfrak{S}_{\mathbf{a}}} t^{n-1-\text{des}(w)}}{(1-t)^{n+1}}. \quad (18)$$

Proof. For each letter e , choose a_e distinct levels from $\{0, 1, \dots, m\}$. At each level, list its selected letters in strictly *decreasing* order, then concatenate the levels in increasing order. This produces a word w of content $\mathbf{a}$, together with a weakly increasing sequence of levels attached to its positions.

Two adjacent positions may share a level only when they are a strict descent in w . Thus a strict increase of levels is required at precisely the $d = n - 1 - \text{des}(w)$ non-descent positions. Conversely, such a level assignment reconstructs the subsets of levels: within an equal-level block the letters are strictly decreasing, so no letter occurs there twice.

Subtract, from each level, the number of required strict increases before that position. This leaves a weakly increasing n -tuple with entries in $\{0, \dots, m - d\}$. Its number is $\binom{m-d+n}{n}$ when $m \geq d$ and zero otherwise. Summing over m yields

$$\sum_{m \geq d} \binom{m-d+n}{n} t^m = \frac{t^d}{(1-t)^{n+1}}.$$

Finally sum over $w \in \mathfrak{S}_{\mathbf{a}}$. □

Theorem 5.3 (Conjecture 4.4 of [1]). *Let $\mathcal{W}_\tau$ be the words of length n on the totally ordered alphabet E for which every order ideal I of τ contributes at least $|I|$ letters. Then*

$$h^*(Q_\tau, t) = \sum_{w \in \mathcal{W}_\tau} t^{n-1-\text{des}(w)}. \quad (19)$$

Proof. A content vector $\mathbf{a}$ of total sum n belongs to B_{τ^*} if and only if $a(F) \leq |F|$ for every filter F of τ . Complementing F converts these inequalities into $a(I) \geq |I|$ for every ideal I . Hence $\mathcal{W}_\tau = \bigsqcup_{\mathbf{a} \in B_{\tau^*}} \mathfrak{S}_{\mathbf{a}}$. Sum (18) over B_{τ^*} , use (17), and compare the result with (6). □

5.3 Extremal coefficients, volume, and zeta evaluation

The proof immediately gives

$$h_0^*(Q_\tau) = 1, \quad h_n^*(Q_\tau) = 0, \quad h_{n-1}^*(Q_\tau) = \#B_{\tau^*}. \quad (20)$$

For the constant coefficient, a word with $n - 1$ descents is the unique strictly decreasing listing of all n letters. For the coefficient at $n - 1$, each admissible content has exactly one weakly increasing word. The total number of admissible words is

$$n! \text{Vol}(Q_\tau) = \sum_{\mathbf{a} \in B_{\tau^*}} \frac{n!}{\prod_{e \in E} a_e!}. \quad (21)$$

This follows either from the leading coefficient in (17) or by setting $t = 1$ in the numerator.

The absence of interior lattice points in Q_τ can also be seen directly: such a point would have every coordinate at least one, while strictness of the inequality for E would force their sum below n . The point $\mathbf{1}$ is interior to $2Q_\tau$. Thus the smallest positive dilation with an interior lattice point is exactly two, consistently with $\deg h^* = n - 1$.

For $k \geq 1$, count the k -multichains in P_τ by their top element. Each coordinate gives a weakly increasing sequence ending at a_e , so

$$Z(P_\tau, k + 1) = \sum_{\mathbf{a} \in P_\tau} \prod_{e \in E} \binom{k + a_e - 1}{a_e}. \quad (22)$$

This equation defines its polynomial continuation. Comparing with (12) at $s = 0$ gives $Z(P_\tau, t) = L_{Q_{\tau^*}}(t - 1)$.

Corollary 5.4 (Ordinary part of Conjecture 4.9). *For every preorder on n elements,*

$$Z(P_\tau, -1) = (-1)^n \# \text{Max}(P_\tau). \quad (23)$$

Equivalently, $\#(\text{int}(2Q_{\tau^}) \cap \mathbb{Z}^E) = \# \text{Max}(P_\tau)$.*

Proof. Apply (17) to τ^* at $m = -2$. Since $\binom{-1}{a_e} = (-1)^{a_e}$ and every base has sum n ,

$$Z(P_\tau, -1) = L_{Q_{\tau^*}}(-2) = \sum_{\mathbf{a} \in B_\tau} \prod_e \binom{-1}{a_e} = (-1)^n \# B_\tau.$$

Use Lemma 2.1 for the first assertion and (5) for the second. $\square$

The source's refined conjecture assigns powers of q to covers of maximal elements. None of the preceding equalities transports that statistic. The ordinary specialization therefore must not be reported as a proof of the full refined conjecture.

6 Colored arrays and matrix Ehrhart–zeta duality

The preceding reciprocity proof is independent of this section. Here the key idea is to replicate *both* color classes in the preorder graph. The two replication counts need not be equal, or even constant over E .

6.1 The heterogeneous enumeration problem

For $\mathbf{p}, \mathbf{q} \in \mathbb{N}^E$, let

$$R_{\mathbf{q}} = \{(e, j) : e \in E, 1 \leq j \leq q_e\}.$$

Define $C_\tau(\mathbf{p}, \mathbf{q})$ as the number of arrays $(x_{e,j})_{(e,j) \in R_{\mathbf{q}}} \in \mathbb{N}^{R_{\mathbf{q}}}$ satisfying

$$\sum_{e \in I} \sum_{j=1}^{q_e} x_{e,j} \leq p(I) \quad (I \in \mathcal{J}(\tau)). \quad (24)$$

Empty groups of coordinates are permitted. The empty array contributes one when all $q_e = 0$.

Aggregating by $a_e = \sum_j x_{e,j}$ gives the explicit finite sum

$$C_\tau(\mathbf{p}, \mathbf{q}) = \sum_{\substack{\mathbf{a} \in \mathbb{N}^E \\ a(I) \leq p(I) \ (I \in \mathcal{J}(\tau))}} \prod_{e \in E} \binom{q_e + a_e - 1}{a_e}. \quad (25)$$

For $q_e = 0$, the factor is one for $a_e = 0$ and zero for $a_e > 0$, as it should be. The ideal E bounds $|\mathbf{a}|$ by $|\mathbf{p}|$, so the sum is finite.

Theorem 6.1 (Heterogeneous array duality). *For every finite preorder and all nonnegative integer vectors $\mathbf{p}, \mathbf{q}$,*

$$C_\tau(\mathbf{p}, \mathbf{q}) = C_{\tau^*}(\mathbf{q}, \mathbf{p}). \quad (26)$$

We prove this by identifying the two sides with the lattice count and the draconian count of one bipartite graph. The graph-duality phenomenon itself is classical in Postnikov's theory [2, Corollary 11.8]; the point here is the exact encoding of the preorder constraints and of two independently varying multiplicity vectors.

6.2 The copied graph and its polytope

Let the left vertices be

$$L = \{0_L\} \sqcup \{(e, i)_L : e \in E, 1 \leq i \leq p_e\}$$

and the right vertices be

$$R = \{0_R\} \sqcup \{(f, j)_R : f \in E, 1 \leq j \leq q_f\}.$$

Join 0_L to every right vertex and every left vertex to 0_R . For the other vertices, put an edge precisely when

$$(e, i)_L \sim (f, j)_R \iff e \leq_\tau f. \quad (27)$$

Denote this graph by $G_\tau(\mathbf{p}, \mathbf{q})$. Its left neighborhoods are nonempty because they all contain 0_R , and the neighborhood of 0_L is all of R .

Let K be the projection, dropping coordinate 0_R , of

$$\sum_{\lambda \in L \setminus \{0_L\}} \Delta_{N(\lambda)}. \quad (28)$$

Equivalently, with $S_e = \{(f, j) : e \leq_\tau f, 1 \leq j \leq q_f\}$,

$$K = \sum_{e \in E} p_e \Delta_{S_e} \subseteq \mathbb{R}^{R_q}.$$

The coefficient of the full simplex indexed by 0_L is zero.

Lemma 6.2. *The polytope K is precisely the nonnegative real solution set of (24). Hence $\#(K \cap \mathbb{Z}^{R_q}) = C_\tau(\mathbf{p}, \mathbf{q})$.*

Proof. Any summand associated with e can contribute to coordinates with labels in an ideal I only if $e \in I$. This proves that points in K satisfy (24).

For the converse, repeat the support-function proof of Proposition 3.1, now on copied coordinates. Given nonnegative coefficients $c_{f,j}$, take a level set of copied coordinates and let T_t be the set of their original labels. Its downward closure $D_\tau(T_t)$ is an ideal. The sum of the selected coordinates is at most the sum of all coordinates with labels in this ideal, and hence at most $p(D_\tau(T_t))$. Integration over levels bounds the functional by

$$\sum_{e \in E} p_e \max(\{c_{f,j} : e \leq_\tau f, 1 \leq j \leq q_f\} \cup \{0\}).$$

This is the support function of K . The extra 0 handles empty neighborhoods after projection. Downward closure handles negative coefficients, and convex separation completes the proof, just as before. $\square$

The unprojected polytope in (28) lies in the hyperplane with coordinate sum $|\mathbf{p}|$. Projection preserves integer points: the missing coordinate is $|\mathbf{p}| - \sum_{(f,j) \in R_q} x_{f,j}$. Thus Theorem 2.2 can count K .

6.3 Reading the draconian inequalities on the other side

Apply (8) with coefficient zero at 0_L and coefficient one at every other left vertex. Every summand in that formula equals one:

$$\binom{\alpha_0}{\alpha_0} = 1, \quad \binom{1 + \alpha_\lambda - 1}{\alpha_\lambda} = 1.$$

Consequently,

$$C_\tau(\mathbf{p}, \mathbf{q}) = \#\{G_\tau(\mathbf{p}, \mathbf{q})\text{-draconian vectors}\}. \quad (29)$$

Let $Q = |\mathbf{q}|$, so that the right color class has $Q + 1$ vertices. A draconian vector has coordinates α_0 and $\alpha_{e,i}$, with

$$\alpha_0 + \sum_{e \in E} \sum_{i=1}^{p_e} \alpha_{e,i} = Q. \quad (30)$$

For a nonempty set A of non-root left vertices, let S be its set of original labels. Its neighborhood has size $1 + q(U_\tau(S))$, so its inequality is

$$\alpha(A) \leq q(U_\tau(S)). \quad (31)$$

These inequalities are equivalent to

$$\sum_{e \in F} \sum_{i=1}^{p_e} \alpha_{e,i} \leq q(F) \quad (F \text{ an order filter of } \tau). \quad (32)$$

Here is a detailed check, including zero multiplicities. Given a filter F , take A to be all the existing left copies with labels in F . If A is empty, (32) is automatic. Otherwise its label set S satisfies $U_\tau(S) \subseteq F$, and (31) implies (32). Conversely, for an arbitrary nonempty A , its coordinates are a subset of all left copies whose labels lie in the filter $U_\tau(S)$. Nonnegativity and (32) imply (31).

Sets of left vertices containing 0_L have the whole right color class as neighborhood, so their inequalities follow from (30). The filter E in (32) makes

$$\alpha_0 = Q - \sum_{e,i} \alpha_{e,i}$$

nonnegative and determines it uniquely. Therefore the draconian vectors are in bijection with nonnegative arrays on the p_e left copies satisfying (32). Since filters of τ are ideals of τ^* , these arrays are exactly those counted by $C_{\tau^*}(\mathbf{q}, \mathbf{p})$. Combined with (29), this proves Theorem 6.1.

6.4 Returning to multichains

Let $\nabla_\tau(k, \ell)$ denote the number of k -multichains in the coordinatewise ordered lattice points of ℓQ_τ . A multichain has k entries; the empty multichain is counted once when $k = 0$.

For $k \geq 1$, convert a multichain $\mathbf{x}^{(1)} \leq \dots \leq \mathbf{x}^{(k)}$ into its coordinate increments:

$$y_{e,1} = x_e^{(1)}, \quad y_{e,j} = x_e^{(j)} - x_e^{(j-1)} \quad (2 \leq j \leq k).$$

These are nonnegative. Since ℓQ_τ is a downset, all the multichain entries are feasible exactly when its last entry is feasible. The latter condition says

$$\sum_{e \in I} \sum_{j=1}^k y_{e,j} \leq \ell |I|.$$

Partial sums provide the inverse map. Thus, including $k = 0$,

$$\nabla_\tau(k, \ell) = C_\tau(\ell \mathbf{1}, k \mathbf{1}). \quad (33)$$

Corollary 6.3 (Conjecture 4.2 of [1]). *For every preorder and all $k, \ell \in \mathbb{N}$,*

$$\boxed{\nabla_\tau(k, \ell) = \nabla_{\tau^*}(\ell, k)}. \quad (34)$$

Proof. Use (33), then Theorem 6.1, then (33) again with τ^* . $\square$

When either parameter is zero, both sides are one. No argument with fractional dilations of a copied preorder is involved. All coefficients in the copied Minkowski sum are zero or one, even though its ambient dimension varies with $\mathbf{p}, \mathbf{q}$.

Self-duality of the preorder is sufficient for symmetry of ∇_τ , but symmetry is not asserted in general. Nor does (26) say that C_τ is jointly a polynomial of bounded degree in the entries of both vectors. Already for a singleton, $C(p, q) = \binom{p+q}{p}$.

7 Worked examples

7.1 Antichains and a single equivalence class

For an n -element antichain, $\mathcal{Q}_\tau = [0, 1]^n$ and $P_\tau = \{0, 1\}^n$. Formula (16) yields

$$\mathbb{E}_\tau(r, s) = \sum_{j=0}^n \binom{n}{j} \binom{s}{n-j} (r+1)^j. \quad (35)$$

Indeed, at a fixed support of size j , the only vector in P_τ with that support has all its nonzero entries equal to one. The heterogeneous array count factors coordinate by coordinate:

$$C_\tau(\mathbf{p}, \mathbf{q}) = \prod_{e \in E} \binom{p_e + q_e}{p_e}, \quad \nabla_\tau(k, \ell) = \binom{k + \ell}{k}^n.$$

Both dualities are visibly symmetric here.

At the other extreme, suppose every two elements of E are equivalent. There is only one nonempty ideal, namely E . Then

$$\mathcal{Q}_\tau(\mathbf{b}; s) = (|\mathbf{b}| + s) \Delta_E, \quad \mathbb{F}_\tau(\mathbf{b}, s) = \binom{|\mathbf{b}| + s + n}{n},$$

and

$$C_\tau(\mathbf{p}, \mathbf{q}) = \binom{|\mathbf{p}| + |\mathbf{q}|}{|\mathbf{p}|}, \quad \nabla_\tau(k, \ell) = \binom{n(k + \ell)}{nk}.$$

These examples confirm that the cardinality n counts elements, not preorder equivalence classes.

7.2 The two-element chain

Let $1 <_\tau 2$. Its inequalities are $x_1 \leq r + s$ and $x_1 + x_2 \leq 2r + s$, with nonnegative coordinates. Direct summation gives

$$\begin{aligned}\mathbb{E}_\tau(r, s) &= \sum_{x_1=0}^{r+s} (2r + s - x_1 + 1) \\ &= \frac{(r + s + 1)(3r + s + 2)}{2}.\end{aligned}$$

After centering,

$$\mathbb{E}_\tau(u - \frac{1}{2}, v - \frac{1}{2}) = \frac{(u + v)(3u + v)}{2},$$

which has the even parity predicted by Corollary 4.4. At $s = 0$, $L_{Q_\tau}(r) = (r + 1)(3r + 2)/2$ and $h^*(Q_\tau, t) = 1 + 2t$.

7.3 A non-self-dual pair of three-element posets

Let V be the poset with $1 < 2$ and $1 < 3$, and let $\Lambda = V^*$. Its two Ehrhart polynomials are

$$L_{Q_V}(m) = \frac{(m + 1)(13m^2 + 17m + 6)}{6}, \quad (36)$$

$$L_{Q_\Lambda}(m) = (m + 1)^2(2m + 1). \quad (37)$$

Consequently,

$$h^*(Q_V, t) = 1 + 8t + 4t^2, \quad h^*(Q_\Lambda, t) = 1 + 8t + 3t^2.$$

Both posets of lattice points have 12 elements, but their Ehrhart and h^* -polynomials differ. This is not a conflict with matrix duality: that theorem exchanges the two parameters as well as the preorder.

In fact the initial matrices, with rows indexed by $k = 0, 1, 2, 3$ and columns by $\ell = 0, 1, 2, 3$, are

$$\nabla_V = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 12 & 46 & 116 \\ 1 & 45 & 436 & 2235 \\ 1 & 112 & 2200 & 20010 \end{pmatrix}, \quad \nabla_\Lambda = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 12 & 45 & 112 \\ 1 & 46 & 436 & 2200 \\ 1 & 116 & 2235 & 20010 \end{pmatrix}. \quad (38)$$

They are transposes, and neither is symmetric. A heterogeneous example is

$$C_V((1, 2, 1), (2, 1, 3)) = C_\Lambda((2, 1, 3), (1, 2, 1)) = 58.$$

For the double polynomial of V , direct enumeration or (12) gives

$$\mathbb{E}_V(u, v) = \frac{(u + v + 1)(13u^2 + 8uv + 17u + v^2 + 5v + 6)}{6}.$$

Its centered form is particularly transparent:

$$\mathbb{E}_V(u - \frac{1}{2}, v - \frac{1}{2}) = \frac{(u + v)(26u^2 + 16uv + 2v^2 + 1)}{12}.$$

Only degrees three and one occur.

There are three maximal elements of P_V : $(0, 1, 2), (0, 2, 1), (1, 1, 1)$. There are four in P_Λ : $(1, 1, 1), (2, 0, 1), (2, 1, 0), (3, 0, 0)$. Thus the respective leading nonzero h^* coefficients are four and three, as (20) predicts. The corresponding zeta evaluations are $Z(P_V, -1) = -3$ and $Z(P_\Lambda, -1) = -4$.

7.4 A mixed preorder with an equivalence class

On $E = \{a, b, c, d, e\}$ let a and c be equivalent, let both lie below e , and impose $b < d$ and $b < e$. No further comparisons are imposed except those forced by reflexivity and transitivity. The exact enumeration gives

$$\begin{aligned} \#P_\tau &= 92, & \# \text{Max}(P_\tau) &= 18, \\ L_{Q_\tau}(m) &= \frac{(m+1)^2(2m+1)(19m^2+21m+6)}{6}, \\ h^*(Q_\tau, t) &= 1 + 86t + 393t^2 + 260t^3 + 20t^4. \end{aligned}$$

The maximal-element count of the dual is 20, while $Z(P_\tau, -1) = -18$. This provides a check on both the dual orientation and the distinction between the two maximal-element counts. The entire double polynomial, in exact Newton-basis form, is supplied in the accompanying JSON data.

8 Exact verification and reproducible artifacts

8.1 Independent enumeration routes

The program code/verify.py uses only the Python standard library and exact integers. It exhausts all reflexive binary relations on $\{0, \dots, n-1\}$ and keeps precisely those satisfying transitivity. Thus it includes all labeled preorders, not merely naturally labeled partial orders or representatives of a few Hasse-diagram shapes.

Lattice points are generated directly from nonnegative vectors whose total sum is bounded by the ideal E , then tested against every nonempty ideal. The program compares this count with both binomial formulas. The word formula is checked by enumerating all n^n words and testing the lower-ideal conditions directly. Maximal elements are found by pairwise coordinatewise comparison, independently of the coordinate-sum criterion.

For matrix and heterogeneous duality, the program evaluates (25) separately for τ and τ^* . This is not an implementation of the graph argument. For interior translation it compares actual finite sets, not only their cardinalities.

8.2 Polynomial checking rather than isolated evaluations

For each preorder, the double polynomial is reconstructed from direct lattice counts at

$$(r, s) \in \mathbb{N}^2, \quad r + s \leq n.$$

In the Newton basis it is

$$\mathbb{E}_\tau(u, v) = \sum_{i+j \leq n} c_{i,j} \binom{u}{i} \binom{v}{j}, \quad (39)$$

where

$$c_{i,j} = \sum_{a=0}^i \sum_{b=0}^j (-1)^{i+j-a-b} \binom{i}{a} \binom{j}{b} \mathbb{E}_\tau(a, b).$$

This triangular grid is unisolvent for polynomials of total degree at most n : order its points by total degree and note that a Newton basis element vanishes at earlier grid points and equals one at its

own point. The reciprocity difference, also of degree at most n , is tested on the translated triangular grid $(r + 1, s + 1)$. For each fixed preorder, these checks certify that difference as a polynomial, given the proved degree bound. They do not replace the proof for arbitrary preorders.

8.3 Recorded run

The supplied run covers 1, 4, 29, 355 labeled preorders of sizes 1, 2, 3, 4, respectively: 389 in total. Its results are:

Check	Number
Direct lattice-count grid points, compared with both formulas	5,642
Unisolvent-grid reciprocity tests	5,642
Interior translation: equality of actual point sets	1,556
Maximal-element characterizations	389
Full h^* vectors versus direct word histograms	389
Ordinary zeta evaluations at -1	389
Matrix duality, $0 \leq k, \ell \leq 3$	6,224
Heterogeneous array duality cases	3,112
Heterogeneous reciprocity cases	3,112

Every check passed. The heterogeneous tests use eight seeded parameter choices per preorder; they are not exhaustive over all multiplicity vectors. Their seed is 20260919. Each nonempty size class is exhausted without random sampling of the relations. The source already reports larger finite checks for some of its conjectures [1, Sections 4 and 7]. Our run is an independent regression test of the formulas and their implementation, not a claim to improve those enumeration bounds.

Run the checks from the package root with

```
python3 code/verify.py --max-n 4 --out data/verification.json
```

The default run took about 19 seconds in the execution environment used for this manuscript. This timing is descriptive, not a benchmark guarantee. The JSON records the measured time, all check totals, and exact data for the worked examples. Repeated runs can differ in timing but should not differ in any mathematical output. The `-max-n 5` option is available but was not run for this report and can be substantially more expensive.

8.4 Complexity and limitations of the checker

The relation generator examines $2^{n(n-1)}$ reflexive relations before filtering for transitivity. For a bound B on the coordinate sum, the raw candidate lattice vectors number $\binom{B+n}{n}$. These are deliberately simple exhaustive algorithms, not efficient methods for large preorders. The exact formulas often give faster counts after P_{τ^*} is available, but the independent brute-force route is useful for detecting sign, orientation, and indexing mistakes.

No floating-point geometric test, numerical root finder, or external computer-algebra system is used in verification. A passing finite run is supporting evidence; the general results rest on the proofs in Sections 3 to 6.

9 Proof audit, boundaries, and further questions

9.1 Dependencies and delicate points

The main dependencies can be summarized without assuming any of the conjectures under investigation:

$$\begin{aligned} \text{Minkowski description} + \text{Postnikov} &\implies \text{polynomial } \mathbb{F}_\tau; \\ \text{interior translation} + \text{Ehrhart reciprocity} &\implies \text{multivariate reciprocity}; \\ \text{reciprocity} + \text{binomial sign identity} &\implies \text{falling formula} \implies \text{word formula}; \\ \text{copied graph} + \text{Postnikov} &\implies \text{heterogeneous duality} \implies \text{matrix duality}. \end{aligned}$$

The classical theorems are stated explicitly and credited. Their proofs are not claimed as new, nor is their full theory reproduced.

Dimension. Interior reciprocity is applied only to a full-dimensional lattice polytope with strictly positive integer parameters. Degenerate boundary parameters are reached by polynomial identity, not by changing the reciprocity sign or invoking the relative interior of a different-dimensional polytope.

The all-ones shift. The amount removed from $x(I)$ is exactly $|I|$, not the number of preorder classes in I . This is why equivalence classes of size greater than one are covered by the same proof. The slack parameter must decrease by one as well; forgetting this shift produces a false formula.

Signs. In deriving the falling formula, the outer sign is $(-1)^n$ and each summand has the same inner sign because $a_0 + \sum_e a_e = n$. The signs cancel only after both are present. In particular, $\mathbb{E}_\tau(m, 0) = (-1)^n \mathbb{E}_\tau(-m - 1, -1)$, with the sign retained when n is odd.

Dual orientation. The neighborhood of a left vertex is an *upper* set. Consequently the associated draconian vectors satisfy upper-set bounds, and therefore lie in the polytope of the *dual* preorder. The non-self-dual example in Equation (38) detects a missing reversal.

Copies and the root coordinate. In the heterogeneous argument the total sum of a draconian vector is $|\mathbf{q}|$, one less than the number of right vertices, not $|\mathbf{p}|$. Its root coordinate is a uniquely determined slack. Zero copies are handled explicitly in the equivalence between subset and filter inequalities.

What computations certify. For each enumerated preorder the exact interpolation checks a whole double polynomial, not just a few unrelated evaluations. No finite list, however, proves the statement for all preorders. That quantifier is handled by the geometric and graph arguments.

9.2 Consequences that should not be overclaimed

A nonnegative word expansion does not by itself imply real-rootedness of h^* : sums of real-rooted polynomials need not be real-rooted without additional structure. Our proof supplies neither

a compatible family nor an interlacing argument. It also does not construct a shelling of the lattice-point poset.

The support enumerator $\sum_{\mathbf{a} \in P_\tau} t^{\#\text{supp}(\mathbf{a})}$ is different from $h^*(Q_\tau, t)$. The source's conjectures about the former must not be inferred from our theorem about the latter. Finally, the q -zeta refinement requires control of a weight statistic that is absent from the unweighted interior and graph counts.

9.3 Concrete directions left by these proofs

The graph proof counts draconian vectors but does not produce a simple, canonical map directly between the two array sets. One can investigate whether a particular triangulation in Postnikov's duality yields a useful explicit algorithm. Such a construction would be stronger than the cardinality identity, even though the latter is now proved here.

The two binomial expansions suggest looking for a direct bijection between the multiset model in (12) and the subset model in (16). A bijection preserving additional statistics could be relevant to the refined zeta question. We do not assume that such a statistic-preserving bijection exists.

The centered-parity identity also provides a smaller space of possible coefficients for double Ehrhart computations. It can be exploited to reduce interpolation costs, although the included checker intentionally does not assume it while reconstructing the polynomial.

10 Conclusion

The selected open problem has a short mechanism: a strict ideal inequality becomes a weak one after subtracting the all-ones vector and reducing the slack by one. Ehrhart reciprocity along integer rays turns that mechanism into the desired identity on the entire parameter space. The same method works with an independent capacity at each element and gives a useful falling-binomial expansion.

A second mechanism, independent copying on the two sides of a bipartite preorder graph, proves the full matrix duality and a heterogeneous strengthening. Together these arguments prove Conjectures 7.2, 4.2, and 4.4 as formulated in [1], and the ordinary specialization of Conjecture 4.9. The proofs have been audited for degenerate parameters, equivalence classes, signs, and reversal of the order, and are accompanied by exact reproducible checks. Their external validation and their priority relative to other work remain separate questions.

A A reusable interior-shift principle

The shortest argument is not specific to preorders. The following statement isolates its scope and also explains why integrality cannot simply be omitted.

Proposition A.1. *Let A be an integer matrix with n columns and no zero rows, and define*

$$K(\mathbf{b}; s) = \{\mathbf{x} \in \mathbb{R}_{\geq 0}^n : A\mathbf{x} \leq A\mathbf{b} + s\mathbf{1}\}.$$

Suppose this family has a polynomial lattice enumerator $F(\mathbf{b}, s)$ for nonnegative integer parameters, and that for all positive integer parameters it is a bounded full-dimensional lattice polytope. Then

$$F(-\mathbf{b}, -s) = (-1)^n F(\mathbf{b} - \mathbf{1}, s - 1)$$

as a polynomial identity.

Proof. Full dimensionality and the absence of zero normals mean that interior points satisfy precisely the strict versions of all the defining inequalities. At integer points, strictness of the upper constraints is $A\mathbf{x} \leq A\mathbf{b} + (s - 1)\mathbf{1}$. Substituting $\mathbf{x} = \mathbf{y} + \mathbf{1}$ yields $A\mathbf{y} \leq A(\mathbf{b} - \mathbf{1}) + (s - 1)\mathbf{1}$, and strict coordinate positivity yields $\mathbf{y} \geq \mathbf{0}$. Apply the same raywise reciprocity and polynomial identity argument as in Theorem 4.2. $\square$

For preorder polytopes, A consists of the indicator rows of nonempty ideals. The substantive additional input is not the translation itself but the lattice-polytope and polynomiality properties, proved using the Minkowski representation. For arbitrary systems of 0–1 inequalities, those properties may fail. Thus the interior-shift calculation alone does not justify a general polynomial identity for every set system.

B Artifact inventory and source audit

The archive contains the editable `article.tex`, its compiled PDF, a build script, the exact verification program, the recorded JSON data and console log, a referee-oriented checklist, and a source-audit note. No external font files or third-party article PDFs are redistributed.

The bibliography cites the public source version used to identify the conjectures, Postnikov’s theorem in its original paper, and Stanley’s classical reciprocity treatment. The source-audit note records the access date and identifies the exact proposition and conjecture numbers relevant to the proofs. The ordinary reciprocal identity and its word consequence are attributed to the source as conjectures; the proofs and weighted extensions in this manuscript are presented separately from those source statements.

References

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